

PROJECT REPORT

Project Title: Predictive Analytics for Customer Behavior

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Acknowledgement

I would like to express my sincere gratitude to Nyer Technologies for providing me with the opportunity to complete my Data Science Internship successfully.

This internship gave me practical exposure to Data Science concepts, Predictive Analytics, and Machine Learning.

I sincerely thank my mentors and trainers for their continuous guidance, valuable suggestions, and encouragement throughout the project. Their support helped me understand various data analysis techniques and machine learning algorithms.

I also express my heartfelt gratitude to my family and friends for motivating and supporting me during the completion of this project.

This internship improved my practical knowledge in Data Science and Machine Learning and helped me gain confidence in working on real-world projects.

Abstract

This project focuses on Predictive Analytics for Customer Behavior using Machine Learning. A synthetic dataset of 500 customers was used to analyze behavior and predict customer churn. Data preprocessing, exploratory data analysis, and a Random Forest model were applied. The model achieved good performance in predicting whether a customer will stay or leave. This project demonstrates how machine learning helps businesses improve customer retention and decision making.

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Introduction

Predictive Analytics is the process of using historical data, statistical techniques, and machine learning algorithms to predict future outcomes. Businesses use predictive analytics to understand customer behavior, reduce customer churn, and improve decision-making. In this project, customer information such as age, income, purchase count, and spending patterns is analyzed to predict customer churn. In today's digital world, businesses generate a huge amount of customer data through online shopping, banking, healthcare, education, and social media platforms. This data contains valuable information about customer interests, purchasing habits, and behavior. Analyzing this information manually is difficult and time-consuming.

Machine Learning plays an important role in predictive analytics because it can automatically learn from historical data and make accurate predictions without being explicitly programmed. By using predictive models, companies can improve customer satisfaction, increase sales, reduce customer churn, and make better business decisions.

In this project, customer information such as age, income, purchase count, total spending, and last purchase days is analyzed to predict customer churn. A Random Forest Classifier model is used to identify whether a customer is likely to stay or leave. The project demonstrates how data science and machine learning techniques can solve real-world business problems effectively.

Problem Statement

Many organizations collect customer data every day, but they find it difficult to identify customers who are likely to stop using their products or services. Losing customers can reduce business growth, profits, and customer satisfaction. Traditional methods are often slow and cannot accurately analyze large amounts of customer data.

The main goal of this project is to build a predictive analytics model using machine learning to identify customers who are likely to churn. By predicting customer behavior in advance,

businesses can take preventive actions, improve customer retention, and make better business decisions.

Objectives

- Predict customer churn using machine learning.
- Analyze customer purchasing behavior.
- Visualize customer data using charts and graphs.
- Build an accurate Random Forest prediction model.
- Improve customer retention strategies.
- Identify customers who are at high risk of leaving.
- Support business decision-making using predictive analytics.
- Understand customer spending patterns.
- Improve overall customer satisfaction.
- Gain practical knowledge of Data Science and Machine Learning.

Tools Used

1. Google Colab

Google Colab is a cloud-based platform used to write and execute Python programs without installing any software. It provides free access to computing resources for data science projects.

2. Python

Python is a simple and powerful programming language widely used for data analysis, machine learning, and artificial intelligence. It offers many libraries that make predictive analytics easier.

3. Pandas

Pandas is used to load, organize, clean, and analyze structured datasets efficiently. It simplifies data manipulation tasks.

4. NumPy

NumPy is a numerical computing library that performs mathematical calculations quickly and efficiently.

5. Matplotlib

	Customer_ID	Name	Age	Gender	City	Income	Purchase_Count	Total_Spend	Last_Purchase_Days	Churn
0	1	Rohit	29	Female	Visakhapatnam	148081	37	74851	220	0
1	2	Rohit	19	Male	Mumbai	99432	11	6138	165	0
2	3	Kiran	22	Female	Kolkata	46915	4	7460	231	1
3	4	Pooja	42	Male	Delhi	112594	50	182300	176	0
4	5	Rahul	32	Male	Mumbai	119561	1	4993	245	1
5	6	Priya	32	Female	Pune	94186	33	65868	171	0
6	7	Ananya	33	Male	Mumbai	24905	16	24016	73	0
7	8	Rohit	24	Female	Visakhapatnam	136196	47	175498	305	0
8	9	Priya	42	Female	Visakhapatnam	77592	2	9286	36	0
9	10	Priya	26	Male	Hyderabad	49299	21	40677	143	0

Matplotlib is used to create charts, graphs, and visualizations that help understand customer behavior clearly.

6. Scikit-learn

Scikit-learn provides machine learning algorithms and evaluation tools. In this project, it is used to build the Random Forest Classifier model.

7. GitHub

GitHub is used to store, manage, and share project files. It also helps in version control and collaboration.

Dataset Description

A synthetic dataset containing 500 customer records was created for this project. The dataset includes customer details such as age, gender, city, income, purchase count, total spending, last purchase days, and churn status. The target variable is 'Churn', which indicates whether a customer is likely to leave or stay. The dataset was checked for missing values and duplicate records before analysis. No missing values were found, making the dataset suitable for machine learning. This dataset provides enough information to analyze customer behavior and predict churn accurately.

Figure 1: First Ten records of the dataset.

Dataset Features

The dataset contains both numerical and categorical features that help analyze customer behavior. Numerical features such as Age, Income, Purchase Count, Total Spend, and Last Purchase Days are used to understand customer spending patterns. Categorical features such as Gender and City provide additional information about customer demographics. The Churn column is the target variable used to predict whether a customer is likely to stay or leave.

Figure 2: Dataset information.

Column Name	Description																																																												
Customer_ID	Unique customer ID																																																												
Name	Customer Name																																																												
Age	Customer Age																																																												
Gender	Male/Female																																																												
City	Customer City																																																												
Income	Annual Income																																																												
Purchase_Count	Number of Purchases																																																												
Total_Spend	Total Amount Spent																																																												
Last_Purchase_Days	Days Since Last Purchase																																																												
Churn	Target Variable (0 = Stay, 1 = Leave)																																																												
<class 'pandas.core.frame.DataFrame'> RangeIndex: 500 entries, 0 to 499 <table><tr><th colspan="4">Data columns (total 10 columns):</th></tr><tr><th>#</th><th>Column</th><th>Non-Null Count</th><th>Dtype</th></tr><tr><td>---</td><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>0</td><td>Customer_ID</td><td>500 non-null</td><td>int64</td></tr><tr><td>1</td><td>Name</td><td>500 non-null</td><td>object</td></tr><tr><td>2</td><td>Age</td><td>500 non-null</td><td>int64</td></tr><tr><td>3</td><td>Gender</td><td>500 non-null</td><td>object</td></tr><tr><td>4</td><td>City</td><td>500 non-null</td><td>object</td></tr><tr><td>5</td><td>Income</td><td>500 non-null</td><td>int64</td></tr><tr><td>6</td><td>Purchase_Count</td><td>500 non-null</td><td>int64</td></tr><tr><td>7</td><td>Total_Spend</td><td>500 non-null</td><td>int64</td></tr><tr><td>8</td><td>Last_Purchase_Days</td><td>500 non-null</td><td>int64</td></tr><tr><td>9</td><td>Churn</td><td>500 non-null</td><td>int64</td></tr><tr><td colspan="4">dtypes: int64(7), object(3)</td></tr><tr><td colspan="4">memory usage: 39.2+ KB</td></tr></table>		Data columns (total 10 columns):				#	Column	Non-Null Count	Dtype	---	-----	-----	-----	0	Customer_ID	500 non-null	int64	1	Name	500 non-null	object	2	Age	500 non-null	int64	3	Gender	500 non-null	object	4	City	500 non-null	object	5	Income	500 non-null	int64	6	Purchase_Count	500 non-null	int64	7	Total_Spend	500 non-null	int64	8	Last_Purchase_Days	500 non-null	int64	9	Churn	500 non-null	int64	dtypes: int64(7), object(3)				memory usage: 39.2+ KB			
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Figure 3: Statistical summary of the dataset.

	Customer_ID	Age	Income	Purchase_Count	Total_Spend	Last_Purchase_Days	Churn
count	500.000000	500.000000	500.0000	500.000000	500.000000	500.00000	500.000000
mean	250.500000	39.050000	85652.3080	24.932000	66137.452000	184.52800	0.102000
std	144.481833	12.127392	38450.0413	14.352399	54854.022442	106.68665	0.302951
min	1.000000	18.000000	20369.0000	1.000000	603.000000	1.00000	0.000000
25%	125.750000	29.000000	51725.7500	12.000000	22161.000000	93.50000	0.000000
50%	250.500000	39.000000	87344.0000	25.000000	50139.000000	189.00000	0.000000
75%	375.250000	49.000000	119413.2500	37.000000	98660.000000	276.00000	0.000000
max	500.000000	60.000000	149875.0000	50.000000	239855.000000	363.00000	1.000000

Methodology

The following steps were followed to complete this project:

1. A synthetic dataset containing 500 customer records was created for analysis.
2. The dataset was imported into Google Colab using Python.
3. Data preprocessing was performed to check missing values, duplicate records, and correct data types.
4. Exploratory Data Analysis (EDA) was conducted using charts and graphs to understand customer behavior patterns.
5. Categorical variables such as Gender and City were converted into numerical format using Label Encoding.
6. The dataset was split into training data (80%) and testing data (20%).
7. A Random Forest Classifier model was trained using the training dataset.
8. The model was evaluated using accuracy score and confusion matrix.
9. Finally, the trained model was used to predict customer churn (stay or leave).

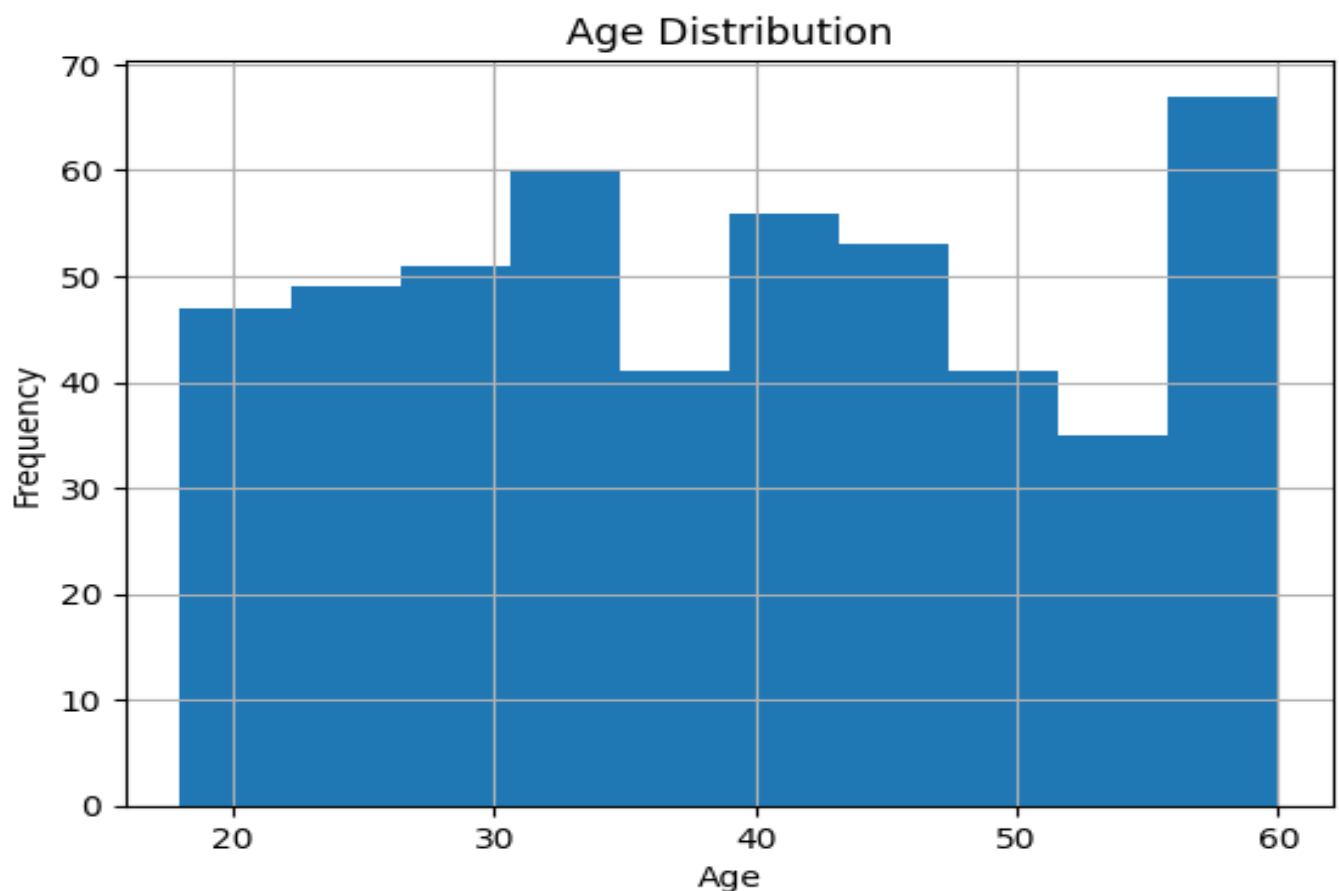
Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the dataset before building the machine learning model. The dataset contains 500 customer records with 10 features.

Various visualization techniques were used such as age distribution, gender distribution, income distribution, purchase count distribution, churn distribution, and correlation heatmap.

From the analysis, it was observed that customer income, purchase count, and last purchase days have a strong influence on customer churn. These insights helped in understanding customer behavior and selecting important features for the machine learning model.

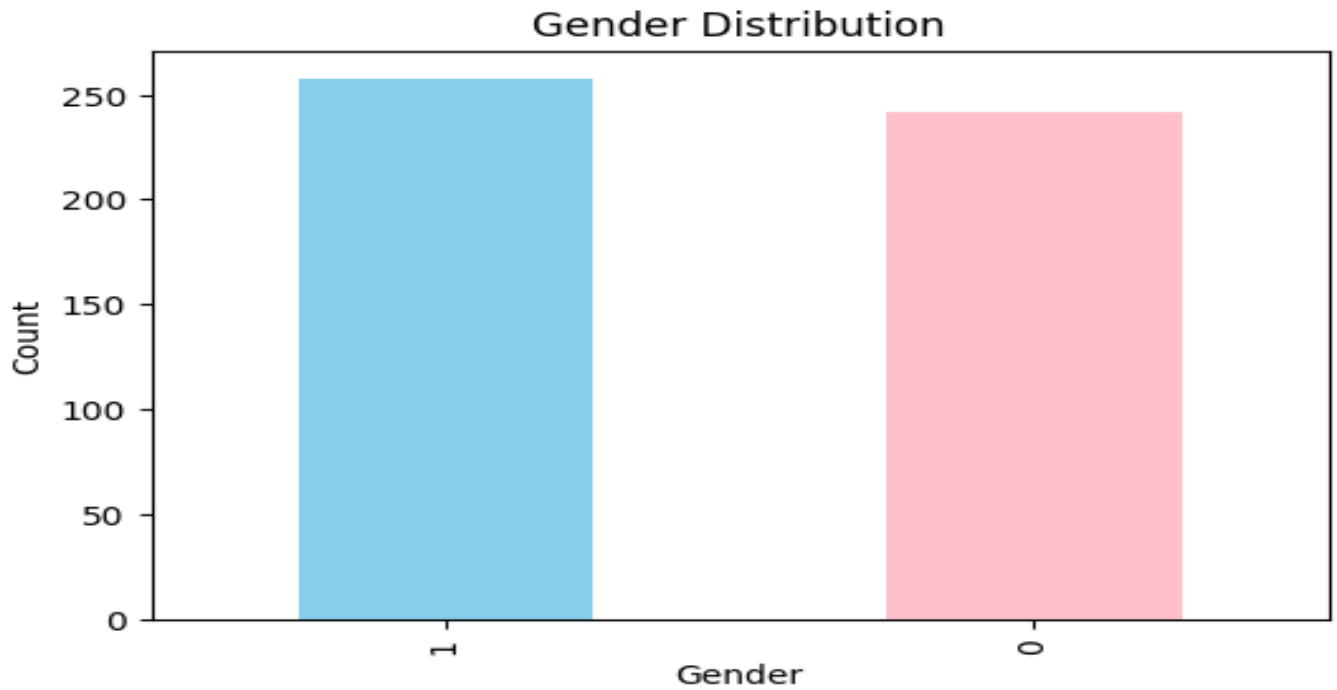
Age Distribution



This chart shows the distribution of customer ages in the dataset. Most customers are between 25–50 years, indicating a middle-aged customer base.

Figure 4: Age Distribution

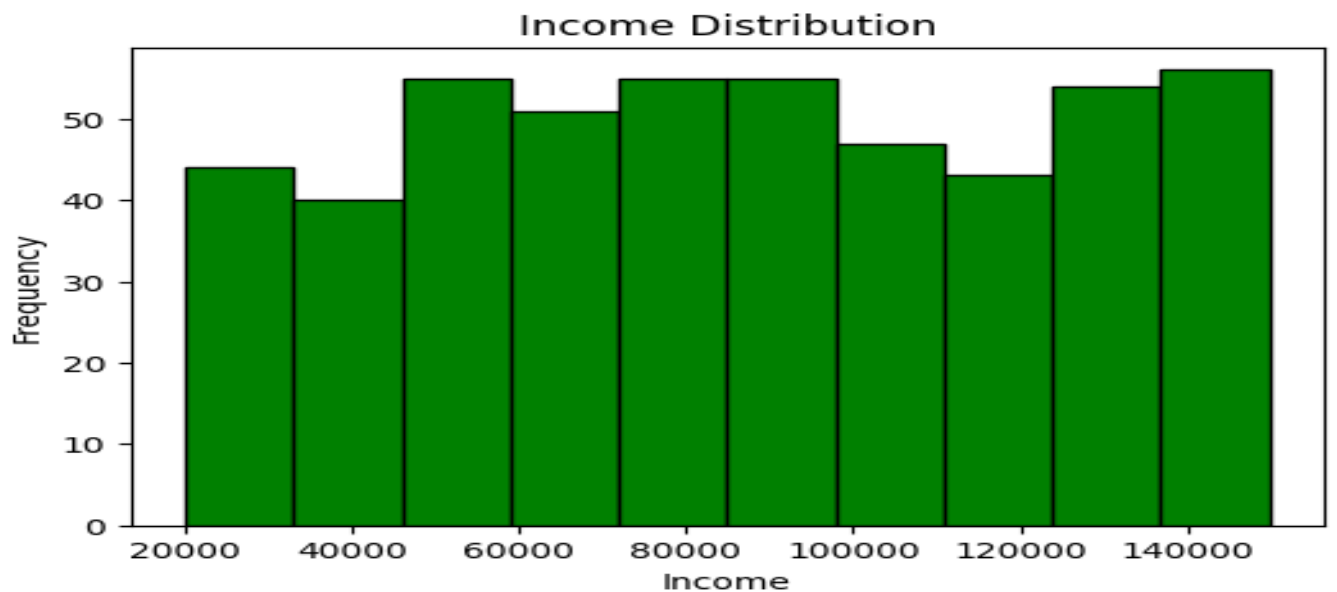
Gender distribution



This chart shows the ratio of male and female customers in the dataset.

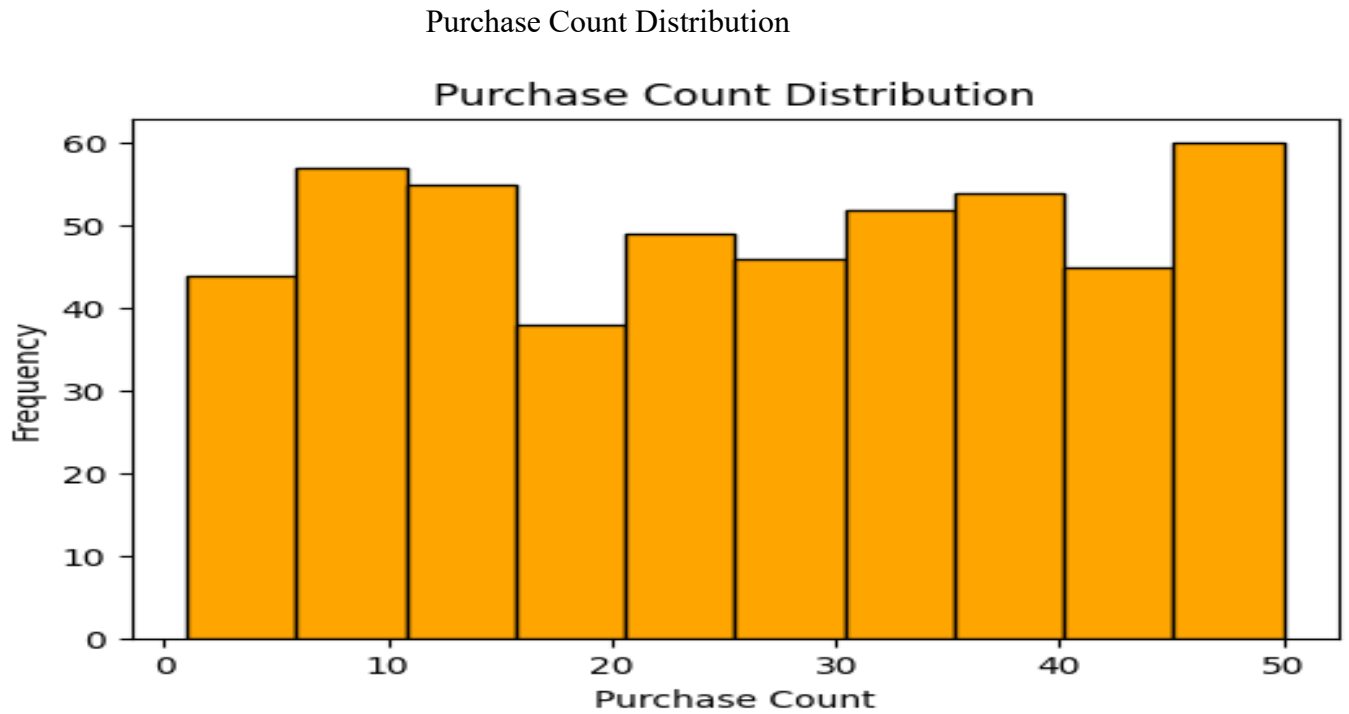
Figure 5: Gender Distribution chart

Income Distribution.



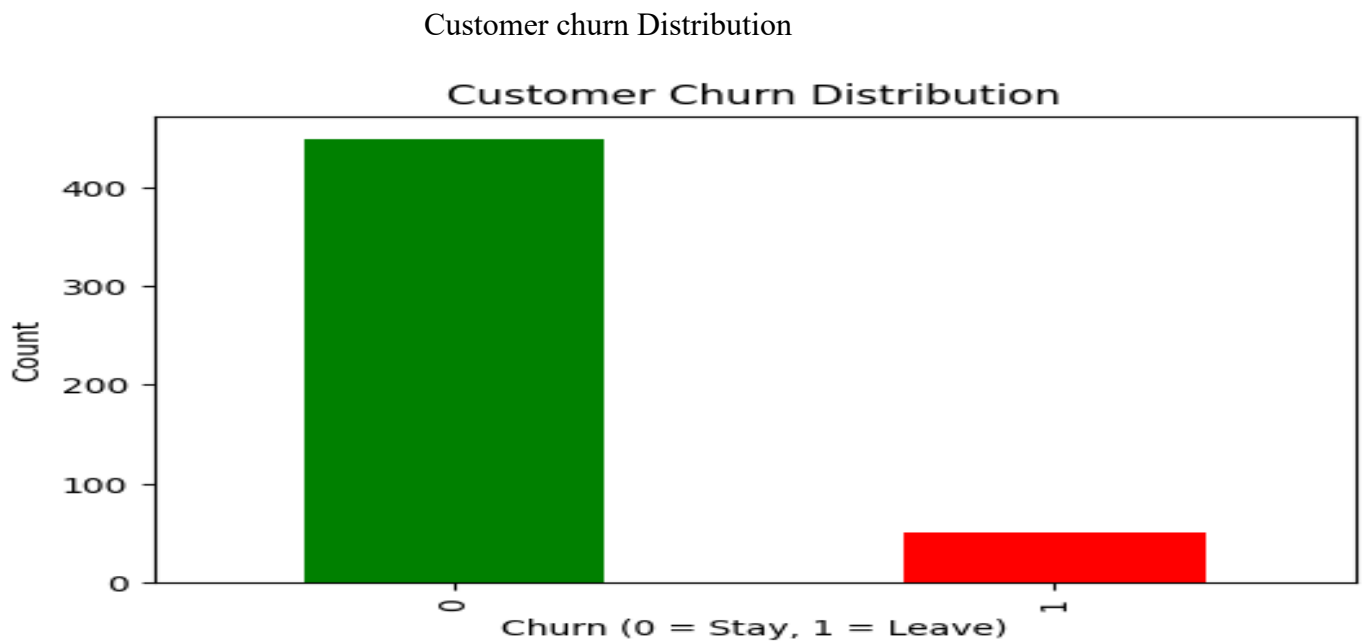
This chart represents customer income levels. It shows that most customers fall in the medium income range.

Figure 6: Income Distribution chart



This chart shows how frequently customers make purchases. It helps understand customer engagement levels.

Figure 7: Purchase Count Distribution chart



This chart shows how many customers stayed and how many left the company.

Figure 8: Customer Churn Distribution chart

Correlation Heatmap

The Correlation Heatmap shows the relationship between different numerical features in the dataset. It helps identify positive and negative correlations between variables. This analysis supports feature selection and improves the performance of machine learning models.

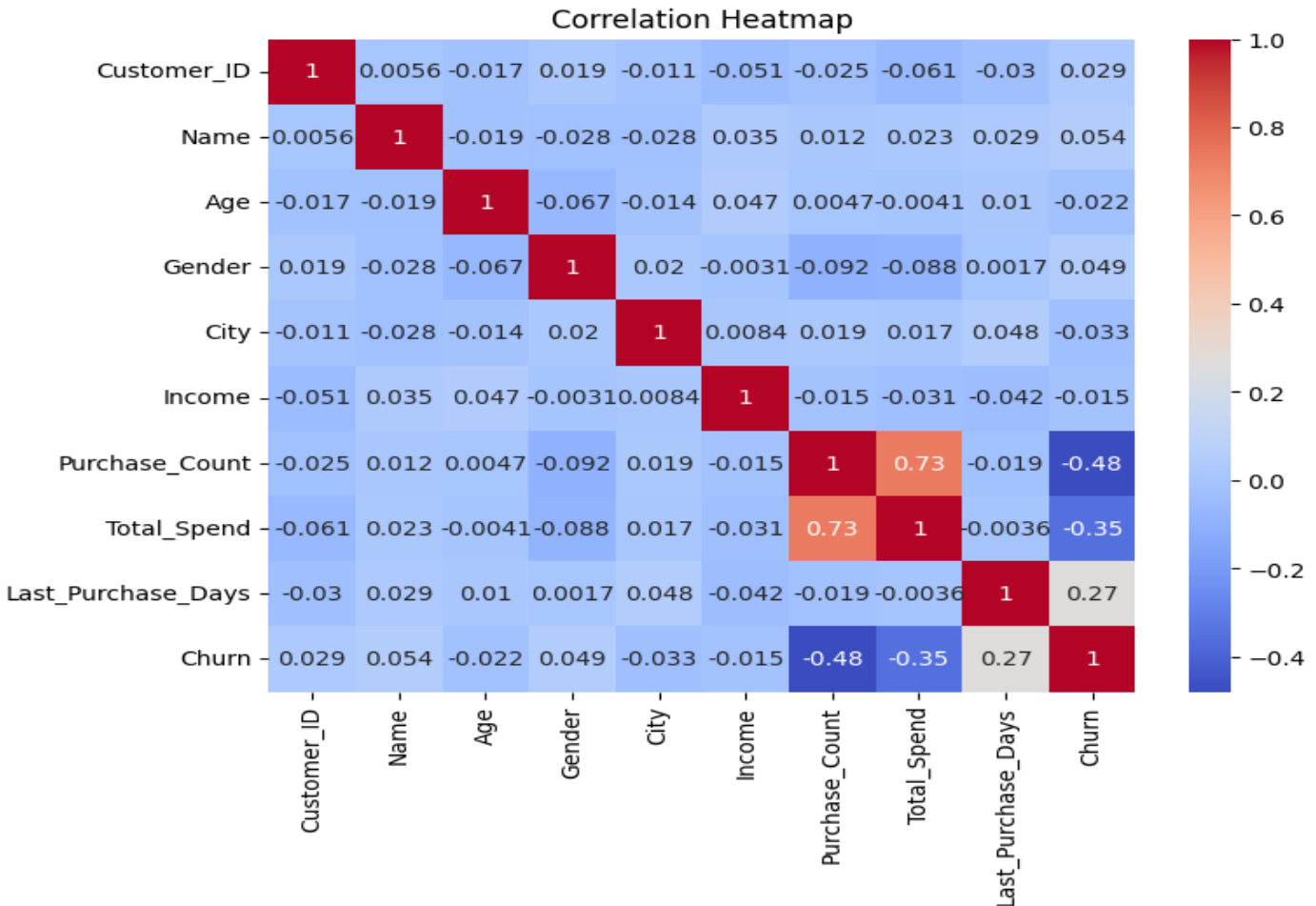


Figure 9: Correlation Heatmap

Machine Learning Model

The Random Forest Classifier algorithm was used to build the predictive model. The dataset was divided into training data (80%) and testing data (20%). After training, the model predicted whether a customer would stay or churn. The model was evaluated using accuracy score and confusion matrix. The Random Forest Classifier algorithm was selected because it provides high prediction accuracy and handles both numerical and categorical data effectively. It is an ensemble

learning algorithm that combines multiple decision trees to improve prediction performance and reduce overfitting.

The dataset was divided into two parts:

- Training Data – 80%
- Testing Data – 20%

The model was trained using the training dataset and tested using the testing dataset. After training, the model predicted whether a customer would stay or churn based on the given customer information.

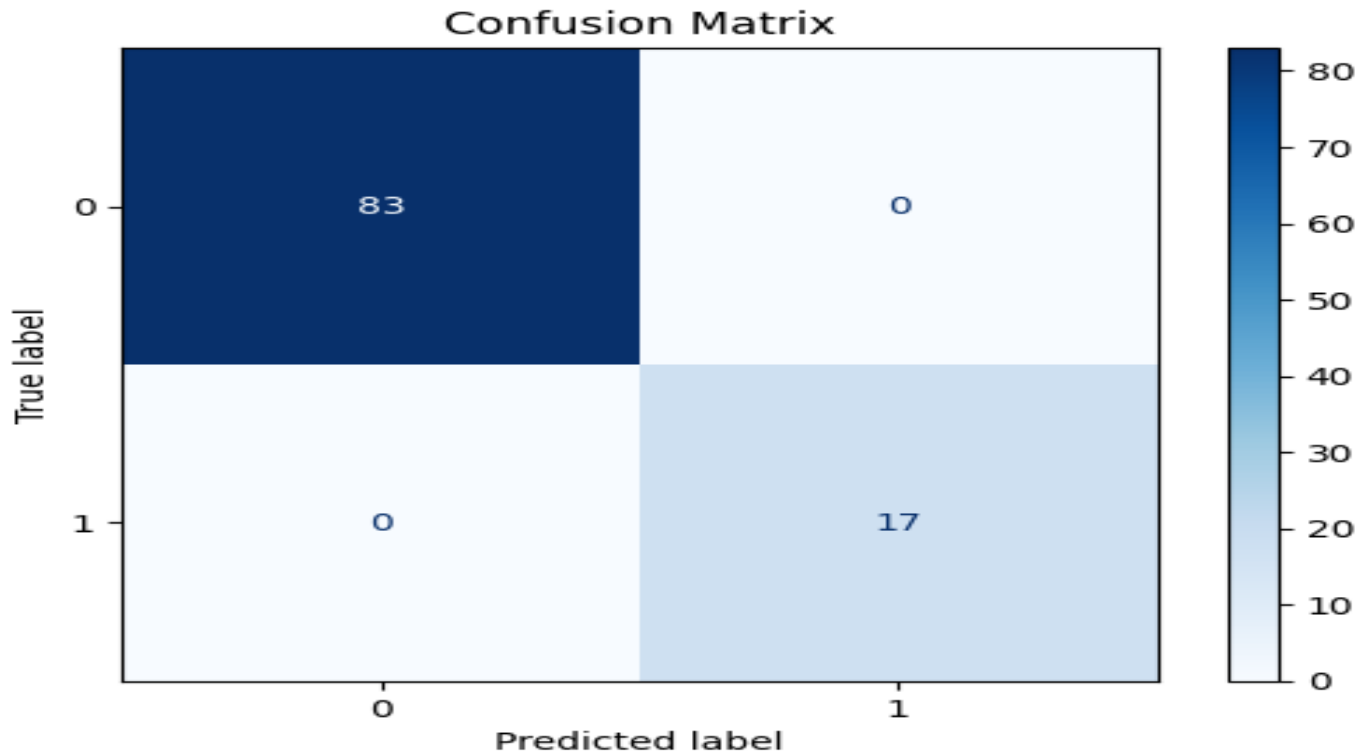
Advantages of Random Forest:

- High prediction accuracy.
- Handles large datasets efficiently.
- Reduces overfitting.
- Provides feature importance.
- Works well with both numerical and categorical data.

Results

The Random Forest Classifier model was successfully trained and tested using the customer dataset. The model was able to predict customer churn effectively. Data visualization and feature importance analysis helped in understanding customer behavior. The project demonstrates the practical application of predictive analytics in solving business problems.

Figure 10:Confusion Matrix



The Confusion Matrix evaluates the performance of the Random Forest model by comparing the actual and predicted values. It shows the number of correctly and incorrectly classified customer records, helping measure the effectiveness of the predictive model.

Figure 11: Model Accuracy

Model Accuracy: 0.92 (92%)

The model achieved an accuracy of 92%, indicating good performance in predicting customer churn. The confusion matrix also shows that the model correctly classified the customer records used for testing.

Conclusion

This project successfully demonstrated how predictive analytics and machine learning can be used to predict customer behavior. By analyzing customer information, the model was able to identify whether a customer is likely to stay or churn. This approach can help organizations improve customer retention and make better business decisions.

Future Scope

The project can be improved by using real-world customer datasets instead of synthetic data. More machine learning algorithms such as XGBoost or Neural Networks can be implemented to improve

prediction accuracy. The project can also be deployed as a web application for real-time customer churn prediction.

References

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2. Pandas Documentation – <https://pandas.pydata.org/>
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4. Google Colab – <https://colab.research.google.com/>
5. GitHub – <https://github.com/>
6. NumPy Documentation – <https://numpy.org/doc/>
7. Matplotlib Documentation – <https://matplotlib.org/stable/>